BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 123 · Defensive Design

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What does authentication establish?

Q2. Why translate high-level source code?
A. The CPU only reads comments
B. It increases physical RAM
C. The CPU needs machine code
D. It removes every logic error

Q3. “Sequence means the computer chooses any random instruction order.” Correct this. Explain why.

Q4. Insertion sort has sorted part [2, 5, 8] and next value 9. Which values move? Give the new sorted part.

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I Do · Read new knowledge

Key vocabulary

Anticipate misuse — Reject unexpected input and avoid exposing sensitive detail in errors.

Input sanitisation — Handle or remove harmful/unexpected characters where appropriate.

Maintainability — Meaningful identifiers, comments and modularity make software easier to inspect.

Defensive design — Plans for invalid data, malicious use and foreseeable user mistakes.

New learning

Anticipate misuse: Reject unexpected input and avoid exposing sensitive detail in errors.

Input sanitisation: Handle or remove harmful/unexpected characters where appropriate.

Maintainability: Meaningful identifiers, comments and modularity make software easier to inspect.

Defensive design: Plans for invalid data, malicious use and foreseeable user mistakes.

Worked example

Improve a login counter that accepts unlimited guesses and prints the correct password on failure.

1. Remove any display of the secret.

2. Limit repeated unsuccessful attempts.

3. Present a non-revealing error and lock/slow further attempts.

4. Keep attempts in an appropriately scoped variable; test boundary conditions.

We Do · Practise with guidance

Guided 1

A text form permits unescaped SQL input. Suggest improvements.

Worked method: Reject unexpected inputs without leaking data.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Complete the condition that locks an account after three failed attempts.

```
attempts = 3
if ... then
    print("Locked")
else
    print("Try again")
endif
```

write

Question 1

State two defensive programming techniques. [2 marks]

Question 2

Explain why meaningful variable names aid maintainability. [2 marks]

Question 3

Explain why showing a user's actual password in an error message is unsafe. [2 marks]

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Exit Ticket · Answer before checking

Input validation is one defensive design technique.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Whether a user has supplied suitable evidence of the claimed identity.

Q2. C — The CPU needs machine code

Q3. Instructions execute in the specified order.

Q4. (a) Values greater than 9 shift right: []. New prefix [2, 5, 8, 9].

I Do / We Do pages · Guided practice

Q1. Reject unexpected inputs without leaking data.

You Do pages · Code First

```
attempts = 3
if attempts >= 3 then
    print("Locked")
else
    print("Try again")
endif
```

Expected result: Locked

You Do pages · Independent questions

Q1. Input validation and restricted access or appropriate sanitisation; any two justified techniques.

Q2. They make purpose easier to understand and reduce debugging/maintenance effort.

Q3. It exposes a secret to observers and can enable unauthorised access.

Exit Ticket · Model answer

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

2. Write a complete new answer in GREEN PEN. Show working if needed: